

Econ 2042: Statistics for Economists

Tutorial Exercise I:

- 1). Describe a sample space that might be appropriate for an experiment in which we roll a pair of dice, one white and one blue.
- 2). With reference to $S = \{1, 2, \dots, 6\}$, in the experiment of rolling a die, describe the event A that the number of points rolled with the die is divisible by 3.
- 3). With reference to $S_1 = \{(x, y) | x = 1, 2, \dots, 6; y = 1, 2, \dots, 6\}$, in the experiment of question 1, describe the event B that the number of points rolled with the pair of dice is 7.
- 4). Use Venn diagrams to show that
 - a). $A \cup (B \cap C)$ is the same event as $(A \cup B) \cap C$.
 - b). $A \cap (B \cup C)$ is the same event as $(A \cap B) \cup (A \cap C)$.
 - c). $(A \cap B)'$ is the same event as $A' \cup B'$.
 - d). $(A \cup B)'$ is the same event as $A' \cap B'$.
 - e). If A is contained in B , then $A \cap B = A$ and $A \cap B' = \emptyset$.
 - f). $(A \cap B) \cup (A \cap B') = A$.
 - g). $(A \cap B) \cup (A \cap B') \cup (A' \cap B) = (A \cup B)$.
 - h). $A \cup (A' \cap B) = (A \cup B)$.
- 5). If $S = \{1, 2, \dots, 9\}$, $A = \{1, 3, 5, 7\}$, $B = \{6, 7, 8, 9\}$, $C = \{2, 4, 8\}$, and $D = \{1, 5, 9\}$, list the elements of the subsets of S corresponding to the following events;
 - a). $A' \cap B$; b). $(A' \cap B) \cap C$; c). $B' \cup C$
 - d). $(B \cup C) \cap D$; e). $A' \cap C$; f). $(A \cup C) \cap D$

- 6). A brewery plans to build a firm in the southern and southwestern parts of Ethiopia. Its management has to decide between sites in Awassa, Shashemene, Arba-Minch, Jimma, Mettu, Gore, Bedelle and Gambella. If " A " represents the event that they will choose a site in Shashemene or Mettu, " B " represents the event that they will choose a site in Awassa or Arba-Minch, " C " represents the event that they will choose a site in Mettu or Gore, " D " represents the event that they will choose a site in Awassa or Mettu, list the elements of each of the following subsets of the sample space, which consists of the eight site selections.

- a). A' ; b). D' ; c). $C \cap D$
d). $B \cap C$; e). $B \cup C$; f). $A \cup B$
g). $C \cup D$; h). $(B \cup C)'$; i). $B' \cap C'$

- 7). A coin is tossed once. Then, if it comes a head, a die is thrown once; if the coin comes up tails, it is tossed twice more. Using the notation in which $(H, 2)$, for example, denotes the event that the coin comes up heads and the die comes up 2, and (T, T, T) denotes the event that the coin comes up tails three times in a row, list:

- a). The elements of the sample space
b). The elements of S corresponding to the event A that exactly one head occurs.
c). The elements of S corresponding to the event B that at least two tails occur or a number greater than 4 occurs.

- 8). An experiment has 4 possible outcomes, A , B , C , D , that are mutually exclusive. Explain why the following assignments of probabilities are not permissible.

- a). $\Pr(A) = 0.12$, $\Pr(B) = 0.63$, $\Pr(C) = 0.45$, $\Pr(D) = -0.20$.
b). $\Pr(A) = 9/120$, $\Pr(B) = 45/120$, $\Pr(C) = 27/120$, $\Pr(D) = 46/120$.

- 9). If we flip a coin twice, what is the probability of getting at least one head?
10). A die is loaded in such a way that each odd number is twice as likely to occur as each even number. Find $\Pr(A)$, where A is the event that a number greater than 3 occurs on a single roll of die.

- 11). A and B are mutually exclusive events; the probability that A occurs is 0.5, and that A or B occurs is 0.6. What is $\Pr(B)$?
- 12). Prove that if B is a subset of A , then $\Pr(A - B) = \Pr(A) - \Pr(B)$.
- 13). In a sample space S , the events A and B are independent. $\Pr(A) = 1/3$, $\Pr(S - B) = 1/4$. Find $\Pr(A \cup B)$.
- 14). 5% of people have high blood pressure. Of the people with high blood pressure, 75% drink alcohol. Of people without high blood pressure, only 50% drink alcohol. What proportion of alcohol drinkers has high blood pressure?
- 15). Three people A , B and C toss a coin in turn until a head occurs. The person who tosses a head first wins. If the coin is biased and head occurs with probability p , what is the probability that B wins?